PROBLEM 12

Conservative Force and Potential III

Original statement · Problems.pdf, p. 9

Oscillation Period

A particle moving under a conservative force oscillates between x_1 and x_2. Show that the period of oscillation is:

$$\tau = 2\int_{x_1}^{x_2} \sqrt{\frac{m}{2[V(x_2) - V(x)]}}\, dx$$

where $V(x)$ is the potential energy function. In particular, if $V(x) = \frac{1}{2}m\omega_0^2(x^2 - bx^4)$, show that the period for oscillations of amplitude a is:

$$\tau = \frac{2}{\omega_0}\int_{-a}^{a} \frac{dx}{\sqrt{a^2 - x^2}\sqrt{1 - b(a^2 + x^2)}}$$

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle of mass m moves on the Ox axis under a conservative force with potential $V(x)$ and oscillates between two turning points $x_1 < x_2$.
- **Constraints:** motion along the axis ($y = z = 0$, smooth guide); $\mathrm{CDOF} = 1$, coordinate x.
- **Hypotheses:** $E = V(x_1) = V(x_2)$, $V(x) < E$ for $x_1 < x < x_2$, and the turning points are simple: $V'(x_1) \neq 0$ and $V'(x_2) \neq 0$. In the particular case, $V(x) = \frac{1}{2} m \omega_0^2 \left(x^2 - b x^4\right)$ and the amplitude is a ($x_1 = -a$, $x_2 = +a$).
- **Parameters:** ω_0 [rad/s], the small-amplitude frequency; b [m^{-2}], the anharmonicity.

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{i}$ along the axis (constant unit vectors).

Derivation

3.1 Kinematics

$$\mathbf{r} = x\,\mathbf{i}, \qquad \mathbf{v} = \dot{x}\,\mathbf{i}, \qquad \mathbf{a} = \ddot{x}\,\mathbf{i}, \qquad \dot{x} = \frac{dx}{dt} \implies dt = \frac{dx}{\dot{x}}$$

3.2 Forces

- **Applied force:** $F = -V'(x)$ (conservative). **Constraint forces:** perpendicular to the axis. **Inertia forces:** none.

3.3 Equations of Motion

- **Newton's second law:** $m\ddot{x} = -V'(x)$. Multiplying by $\dot{x}$, the energy is a first integral:

$$\frac{d}{dt}\left(\tfrac{1}{2} m \dot{x}^2 + V\right) = \dot{x}\left(m\ddot{x} + V'\right) = 0 \implies \tfrac{1}{2} m \dot{x}^2 = E - V(x) \implies \dot{x} = \pm\sqrt{\frac{2}{m}\left[E - V(x)\right]}$$

3.4 Integration and Answers

(a) General formula for the period

Pedagogical Justification: the equation is first order and separable, so the time between two points is an integral. Going from x_1 to x_2 the velocity is positive, $\dot{x} = +\sqrt{\frac{2}{m}\left[E - V(x)\right]}$, and

$$dt = \frac{dx}{\dot{x}} = \sqrt{\frac{m}{2\left[E - V(x)\right]}}\,dx \implies \frac{\tau}{2} = \int_{x_1}^{x_2} \sqrt{\frac{m}{2\left[E - V(x)\right]}}\,dx$$

- On the way back the velocity is negative, $\dot{x} = -\sqrt{\frac{2}{m}\left[E - V(x)\right]}$, and x goes from x_2 to x_1. Reversing the limits cancels the minus sign:

$$\int_{x_2}^{x_1} \frac{dx}{-\sqrt{\frac{2}{m}\left[E - V(x)\right]}} = \int_{x_1}^{x_2} \frac{dx}{\sqrt{\frac{2}{m}\left[E - V(x)\right]}}$$

so the return trip takes the same time (the motion is symmetric under time reversal) and the period is twice the one-way time. Using $E = V(x_2)$:

$$\boxed{\tau = 2\int_{x_1}^{x_2} \sqrt{\frac{m}{2\left[V(x_2) - V(x)\right]}}\,dx} \quad \blacksquare$$

- **Convergence.** The integrand is singular at the end points, where $V(x_2) - V(x) \to 0$. At a simple turning point, $V(x_2) - V(x) \approx -V'(x_2)(x - x_2)$ with $V'(x_2) \neq 0$, so the integrand behaves as $(x_2 - x)^{-1/2}$. Its antiderivative is $-2(x_2 - x)^{1/2}$, which is finite at x_2, so by Barrow's rule the integral converges (and the same at x_1).

(b) Application to $V = \frac{1}{2}m\omega_0^2\left(x^2 - bx^4\right)$

> **Pedagogical Justification:** for the amplitude a the turning points are $\mp a$, so we form $V(a) - V(x)$ and factor the common harmonic term $\left(a^2 - x^2\right)$ with the identity $a^4 - x^4 = \left(a^2 - x^2\right)\left(a^2 + x^2\right)$.

$$V(a) - V(x) = \tfrac{1}{2}m\omega_0^2\left[\left(a^2 - x^2\right) - b\left(a^4 - x^4\right)\right] = \tfrac{1}{2}m\omega_0^2\left(a^2 - x^2\right)\left[1 - b\left(a^2 + x^2\right)\right]$$

- The factor under the root of the general formula is

$$\frac{m}{2\left[V(a) - V(x)\right]} = \frac{m}{m\omega_0^2\left(a^2 - x^2\right)\left[1 - b\left(a^2 + x^2\right)\right]} \implies \sqrt{\cdot} = \frac{1}{\omega_0\sqrt{a^2 - x^2}\sqrt{1 - b\left(a^2 + x^2\right)}}$$

$$\boxed{\tau = \frac{2}{\omega_0}\int_{-a}^{a}\frac{dx}{\sqrt{a^2 - x^2}\,\sqrt{1 - b\left(a^2 + x^2\right)}}} \quad \blacksquare$$

Interpretation & Checks

- **Harmonic limit ($b \to 0$):** the integral is $\int_{-a}^{a}\frac{dx}{\sqrt{a^2-x^2}}$. With $x = a\sin\varphi$, $dx = a\cos\varphi\,d\varphi$, limits $\varphi \in [-\pi/2, \pi/2]$ (where $\cos\varphi \geq 0$, so $\sqrt{a^2 - x^2} = a\cos\varphi$) and Barrow's rule:

$$\int_{-\pi/2}^{\pi/2}\frac{a\cos\varphi\,d\varphi}{a\cos\varphi} = \left[\varphi\right]_{-\pi/2}^{\pi/2} = \pi \implies \tau = \frac{2\pi}{\omega_0}$$

the isochronous harmonic period, independent of the amplitude.

- **Softening ($b > 0$):** the factor $\sqrt{1 - b\left(a^2 + x^2\right)} < 1$ makes the integrand larger than in the harmonic case, so $\tau > 2\pi/\omega_0$. In the form with $x = a\sin\varphi$ (next item) the integrand $1/\sqrt{1 - ba^2\left(1 + \sin^2\varphi\right)}$ increases with a for every φ, so the period grows with the amplitude. For $b < 0$ (hardening) the factor is larger than 1, $\tau < 2\pi/\omega_0$, and every amplitude gives a bounded oscillation.

- **Small anharmonicity:** with $x = a\sin\varphi$ the period reads $\tau = \dfrac{2}{\omega_0}\displaystyle\int_{-\pi/2}^{\pi/2}\frac{d\varphi}{\sqrt{1 - ba^2\left(1+\sin^2\varphi\right)}}$. Expanding to first order in ba^2 ($(1-\epsilon)^{-1/2} \approx 1 + \epsilon/2$) and using $\int_{-\pi/2}^{\pi/2}\sin^2\varphi\,d\varphi = \left[\dfrac{\varphi}{2} - \dfrac{\sin 2\varphi}{4}\right]_{-\pi/2}^{\pi/2} = \left(\dfrac{\pi}{4} - \dfrac{\sin\pi}{4}\right) - \left(-\dfrac{\pi}{4} - \dfrac{\sin(-\pi)}{4}\right) = \dfrac{\pi}{2}$, together with $\int_{-\pi/2}^{\pi/2} d\varphi = [\varphi]_{-\pi/2}^{\pi/2} = \pi$ for the constant term:

$$\tau \approx \frac{2}{\omega_0}\left[\pi + \frac{1}{2}ba^2\left(\pi + \frac{\pi}{2}\right)\right] = \frac{2\pi}{\omega_0}\left(1 + \frac{3}{4}ba^2\right)$$

- **Admissible amplitudes ($b > 0$):** $V' = m\omega_0^2\left(x - 2bx^3\right) = 0$ gives $x^2 = 1/(2b)$, where V has its maximum, so bounded oscillations require $a^2 < 1/(2b)$; as $a^2 \to 1/(2b)$ the factor $1 - b\left(a^2 + x^2\right)$ vanishes at the turning points and $\tau \to \infty$.
- **Dimensional check:** $[b] = \mathrm{m}^{-2}$, so $b\left(a^2 + x^2\right)$ is dimensionless; $\int dx/\sqrt{a^2 - x^2}$ is dimensionless and $[\tau] = 1/[\omega_0] = \mathrm{s}$.
- **Check against the official key:** the key of `Problems.pdf` only gives the hint to separate variables (the period formula is part of the statement). The result is verified numerically ($\omega_0 = b = 1$; quadrature against DOP853 integration of $\ddot{x} = -x + 2bx^3$):

Amplitude a	**Quadrature**	**ODE**	**First-order estimate**
0.1	6.3309	6.3309	6.3303
0.3	6.7592	6.7592	6.7073
0.5	8.0086	8.0086	7.4613
0.7	16.9473	16.9473	8.5923 (outside its range of validity)

Source: Problems.pdf